

has done is to translate a classic material deposition method—spray painting—to a way of manufacturing cheap electronic devices,” says Calvin Chan, an electrical engineer at NIST.

Solar cells based on advanced thin-film absorbers

Using a continuous-flow microreactor, researchers at Oregon State University have developed thin-film absorbers for solar cells that could significantly reduce the cost of solar energy devices as also material waste. The enabling technology is said to be safer, faster and more economical than previous chemical solution approaches, and could be used to continuously and rapidly deposit thin-film absorbers for solar cells from such compounds as copper indium diselenide.

Copper indium diselenide is one of the leading photovoltaic alternatives to silicon-based solar energy devices. The earlier approaches to its use have relied on such methods as sputtering, evaporation and electrodeposition, which can be time-consuming, or require expensive vacuum systems or exotic chemicals that raise production costs.

However, the new technology to deposit nanostructure films on various surfaces in a continuous-flow microreactor addresses some of these issues

and makes the use of this process commercially practical.

‘Glasses-free’ 3-D technology

From scientists to companies, everybody seems to be exploring the possibilities of developing a 3-D technology that requires no glasses. Japanese companies are doing their bit to develop this technology at affordable prices.

A ‘glasses-free’ 3-D technology produced by a Japanese company sends different signals to each eye, creating the illusion of 3-D images. According to a company representative, future developments will lead to sharper images and less eye fatigue.

At present, a 165cm (65-inch) set based on this technology costs around \$32,000—about three times more than the price of a 2-D television of the same size. There are also problems for people with visual impairment. If one eye is stronger than the other, the 3-D image will suffer.

Side effects of 3-D TV viewing

Samsung has warned 3-D TV users of an array of side effects on constant exposure to this medium. It has highlighted potential dangers that the technology poses to pregnant women, the elderly, children and people with serious medical conditions.

The devices could trigger epi-

leptic fits or cause ailments ranging from altered vision and dizziness to nausea, cramps, convulsions and involuntary movements such as eye or muscle twitching, according to the company. Those who have been deprived of sleep or drinking alcohol are also advised to avoid watching 3-D television.

The scale of the warning threatens to overshadow the launch of 3-D, which has been billed as the future for television by manufacturers and broadcasters.

New ultrasound imaging technology

California-based Samplify Systems has announced an auto-focus beamforming technology that uses integrated Auto-Focus engine to automatically refocus the receiver to capture reflections at different scan depths. The system thereby alleviates the real-time calculation requirement conventionally done in software and can simplify software and system design.

The technology uses a 32-channel ultrasound analogue front-end receiver module in an ultra-small dual-in-line configuration based on the SAM1600 family of compressing analogue-to-digital converters.

Four scan-lines are formed simultaneously using phased array technology for higher frame rates and improved

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